

EARLY-STAGE CONCEPT · OPEN FOR FEEDBACK

AI-Powered Elderly Care Monitoring System

A privacy-first, smartphone-powered surveillance concept — presented for early review and group discussion.

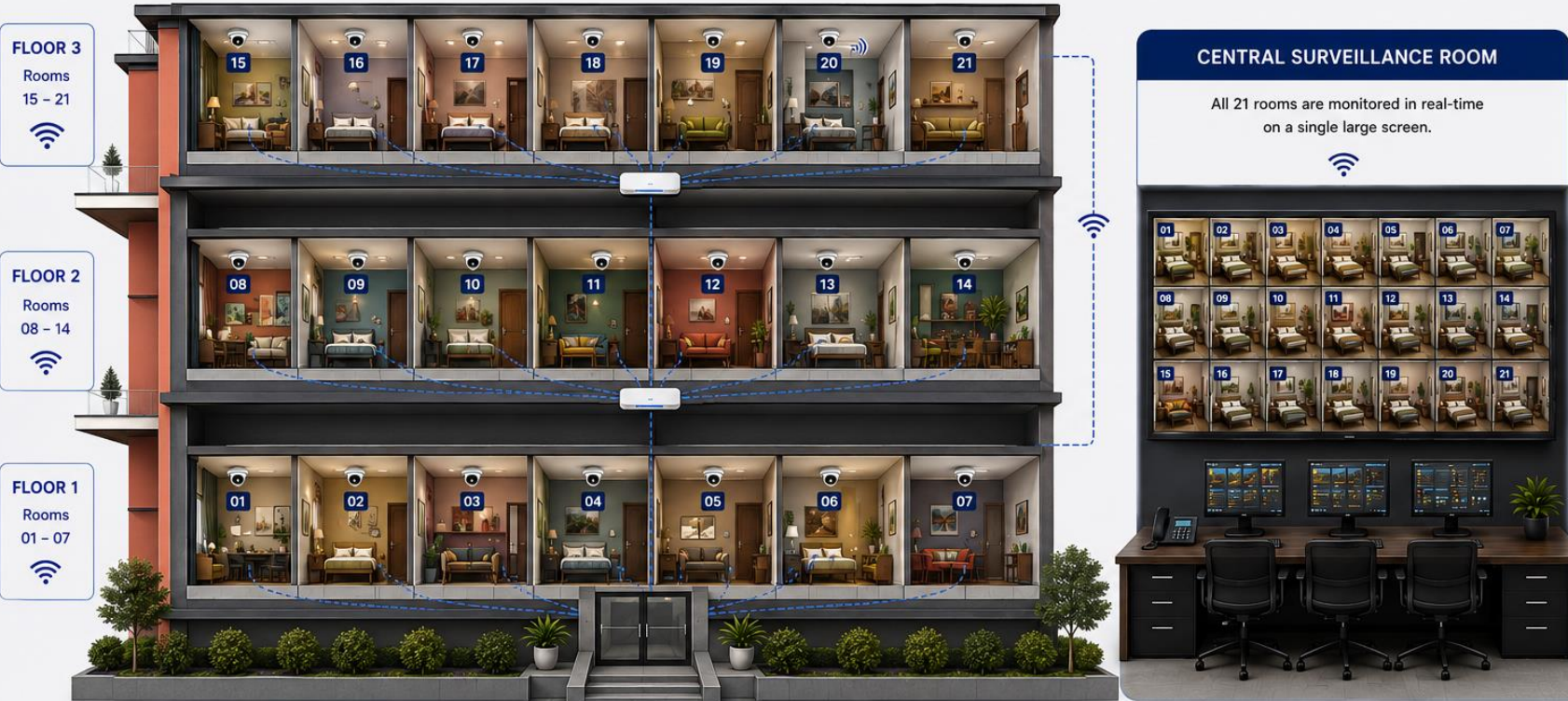
Overall System Overview

1

OVERALL SYSTEM OVERVIEW

21 Rooms Connected Over Secure Wi-Fi Network

All data streams securely to the central surveillance room.



21 Rooms
Across 3 Floors



360°
Coverage



AI-Powered
Monitoring



Real-Time
Alerts



Centralized
Monitoring



Secure &
Private



Scalable
Solution

Room-Level Hardware Setup

2

ROOM-LEVEL HARDWARE SETUP

Each room is equipped with a 360° fisheye camera connected to a smartphone for local AI processing.



360° FISHEYE CAMERA

- Ceiling-mounted (facing down)
- 360° panoramic view
- Full room coverage with a single camera
- High-resolution video capture



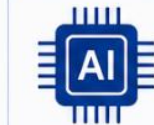
SMARTPHONE (EDGE AI PROCESSOR)

- Mounted flat on the ceiling
- Receives video feed from camera
- Performs local AI inference
- Detects events in real-time
- Stores temporary data



POWER FROM CEILING LIGHTING

- Powered from existing ceiling lighting circuit
- Single cable routing inside the ceiling
- 24/7 uninterrupted operation
- No visible power outlet or wiring in the room



LOCAL PROCESSING

- All AI processing is done locally
- No raw video leaves the room
- Ensures privacy and reduces bandwidth usage



All-in-One Installation

All devices are ceiling-mounted at a single location for a clean, minimal, and unobtrusive setup.



Private & Secure

All processing happens inside the room. No raw video is shared.



360° Full Coverage

One camera covers the entire room with no blind spots.



Low Maintenance

Minimal wiring and easy to maintain.



Cost-Effective

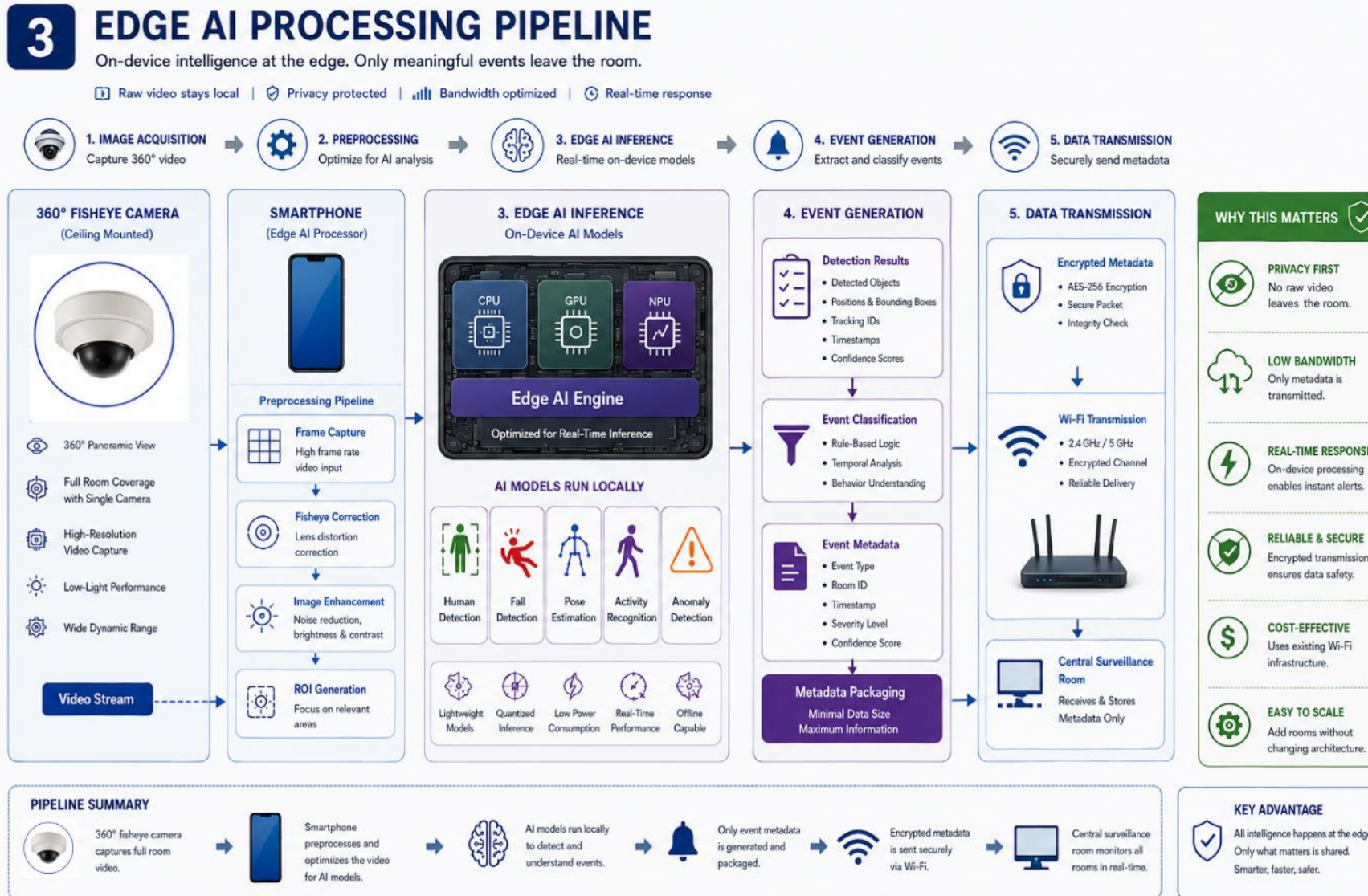
Leverages smartphones for lower hardware and deployment costs.



Energy Efficient

Low power, 24/7 operation using existing ceiling infrastructure.

Edge AI Processing Pipeline



Why Smartphone Instead of Embedded Board?

4 WHY SMARTPHONE INSTEAD OF EMBEDDED BOARD?

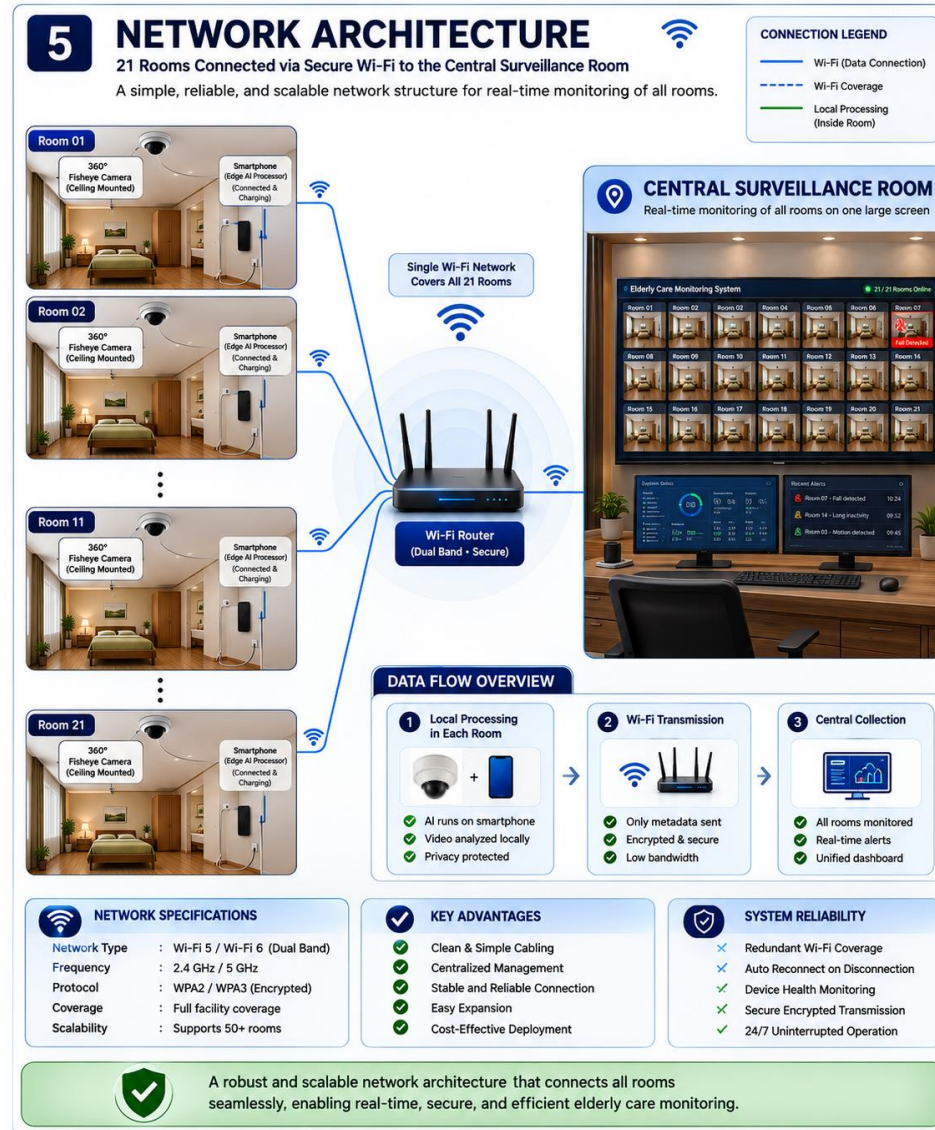
Smartphone-based edge AI delivers better performance, lower cost, and easier deployment.

TRADITIONAL EMBEDDED BOARD	VS	SMARTPHONE-BASED EDGE COMPUTING	✓
 Higher hardware cost Specialized board + AI accelerator	 Hardware Cost	 Lower overall system cost Utilizes mass-market smartphone economies of scale	✓
 Lower computational power Limited by embedded hardware	 Performance	 High-performance CPU / GPU / NPU Delivers superior AI inference performance	✓
 Limited upgradeability Hardware is fixed and difficult to upgrade	 Upgradability	 Easy software updates Continuous improvements via OS / app updates	✓
 Dedicated AI accelerator required Increases complexity and cost	 AI Capability	 Built-in AI acceleration NPU/GPU available in modern smartphones	✓
 Complex integration and maintenance Requires hardware design expertise	 Integration & Maintenance	 Simpler integration & maintenance Standard devices, easy replacement and support	✓
 Higher development and deployment cost Longer development cycle	 Development & Deployment	 Faster development & deployment Leverage existing smartphone ecosystem	✓
 External power management often needed Additional components required	 Power Management	 Efficient power management Built-in battery management and power efficiency	✓
 Higher total cost of ownership Hardware, accessories, and labor	 Total Cost of Ownership	 Lower total cost of ownership Lower hardware, maintenance, and operational cost	✓



Smartphone-based solution is cost-effective, powerful, scalable, and easy to maintain.

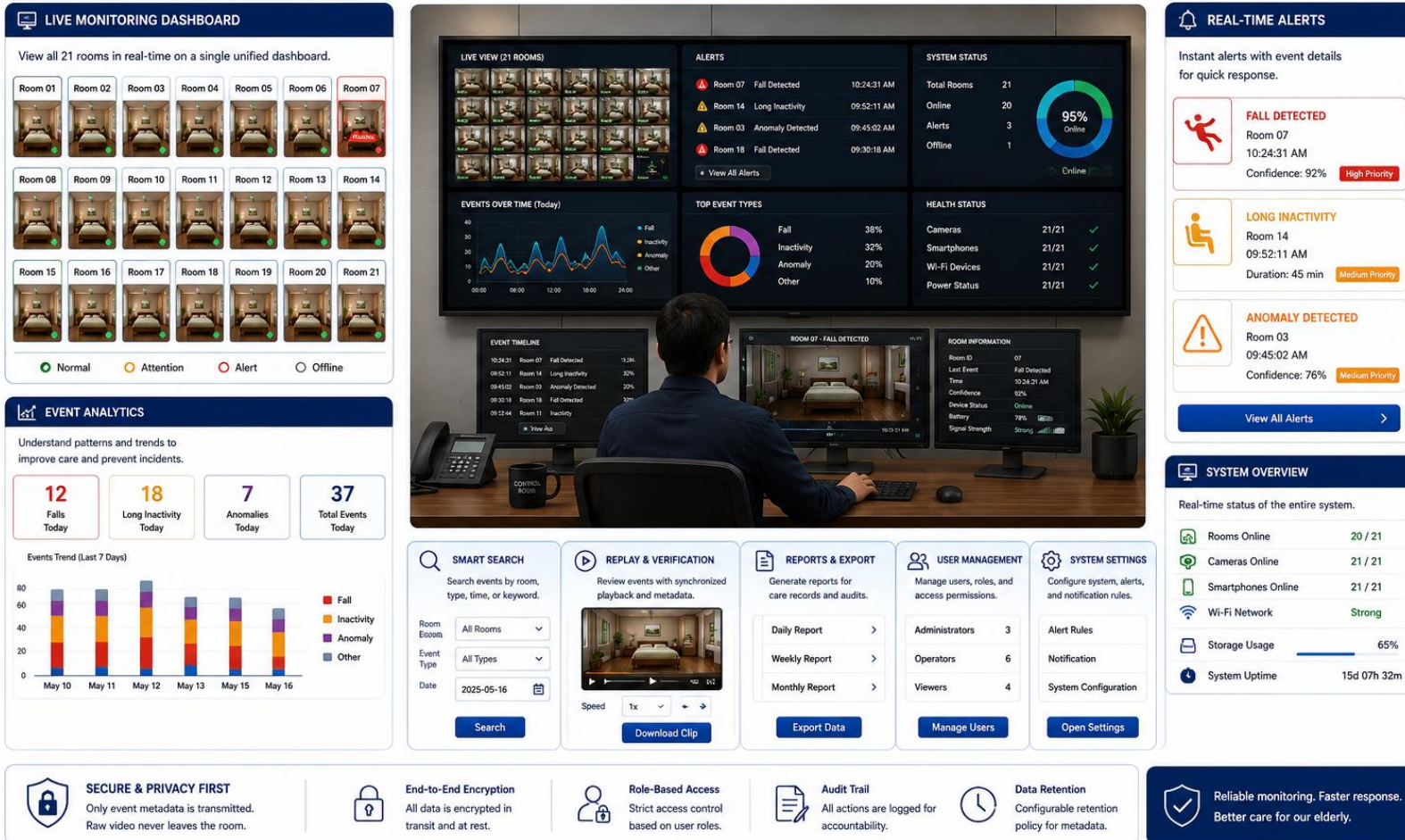
It enables high-performance AI at the edge with lower cost, faster deployment, and future-ready scalability.



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SURVEILLANCE CONTROL ROOM

Centralized monitoring, real-time alerts, and intelligent decision support.



Complete Integrated System



LET'S TALK

Questions, Comments & Feedback

This deck reflects early-stage ideas, not final decisions — every part of it is open to change.

Thank you for your time and input.